SECJ2203: Software Engineering

Project Proposal

KADA Cooperative System

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Faculty of Computing

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1. Introduction

1.1 **The Goal**

Kemubu Agricultural Development Authority (KADA) is the primary agency dedicated to the improvement of the living standards of farmers in the Kemubu region of Malaysia, KADA over the years has managed to control its applications for loans and membership through the use of manual processes, which is inefficient, time-wasting as well as environmentally unfriendly and has poor effects such as loss of records and delays in handling different applications. As a way of addressing these issues, KADA has proposed a new innovative solution, a mobile application labeled the “KADA Cooperative System”, which seeks to increase accessibility and achieve efficiency in the operations by digitizing every process of the loan and membership application.

The objective of this system is to modernize outdated manual processes in order to provide KADA personnel and members with a more reliable, efficient, and secure means of managing cooperative services. KADA will be able to serve its members by modifying these processes which will reduce the time taken to process applications, ensure data safety, and promote environmentally sustainable practices.

The objectives of this project include:

a) Students will be taught the application phases of the SDLC; Planning, Analysis, Design, Implementation and Testing for KADA development on a new member and loan application system in order to achieve the desired goal of the project.

b) Students will evaluate the key features of the new member and loan application proposed system, especially membership registration, loan processing, reporting among the Board of Directors, and finally come up with a final design for the system.

c) In order to ensure that it works efficiently and improves over the time, students will concentrate on the core elements of the system, such as database management, user interface and member-loan management logic.

d) Students would execute testing methodologies to guarantee the running system works smoothly, correct any error(s), and maintain performance solely on member registration and loan application processing.

1.2 The Scope

It is a project targeted at the development of a fully integrated digital platform for KADA's cooperative system in the Kemubu area. Emphasis will be given to replacing manual and paper-based processes involved with membership and loan applications with an efficient and eco-friendly online alternative. The said system should serve to simplify the process of registration and applications, both in terms of membership and loans, while managing data; hence, proffering users a wide interface to apply, manage, and track membership and loan statuses with ease. This includes processes such as user registrations, application submissions, monitoring of the status of the applications in real time, and even storage of data in a secure way for administrators. Various technologies will support the system.

These are the technologies that will be applied in the system:

1) Hypertext Preprocessor (PHP)
2) MySQL

No	Technology	Description
1	Hypertext Preprocessor (PHP)	PHP is used for server-side scripting in the KADA system to build dynamic web content, handle form data of users and ensure compatibility across various operating systems.
2	MySQL	MySQL efficiently manages large databases for the KADA system, handling substantial data on users and their loan details with speed and reliability.

2. Software Process Model

2.1 The Goal

A software process model aims to serve the various tasks in a controlling and coordinating perspective so as to realize the final output as well as the system to be delivered to the stakeholders. This software process model should fit the needs of the KADA cooperative system by changing the traditional system that still requires filling of user information using papers to our new system which allows doing applications online and running the cooperative better. The software process model that will be used in this system should define all the segments of the work being done, and specify all the possible errors that can be avoided from occurring all the way to actual implementation.

A properly structured software process model is beneficial for the team as it promotes the understanding of the role of each individual within the task of the project, and ensures the project remains on track and all team members are up to date with progress made. Thus, the software process model is defined as following:

- Clear responsibilities and roles for each team member.
- Identify the potential risks at early stages.
- The stage of development for each phase.
- The tasks and steps to be completed within each phase.
- The process for each task meets the requirements.

2.2 The Chosen Software Process Model

The Waterfall Model is the best choice for developing the KADA System due to its structured and linear approach which can make it easier for developers to understand each phase sequentially. This model is organized into clearly defined steps. Each step must be completed before moving to the next step. The clear instructions and steps, the Waterfall Model helps the team to stay focused on achieving the objectives without distraction. Additionally, the model prioritizes through documentation which is essential for tracking project details and ensuring team alignment. If there are any new changes identified, the team can catch up and do the adjustment with their project to prevent any mistakes from happening.

Figure 2.1 illustrates the six stages in the Waterfall Model which is requirement gathering and analysis, system design, implementation, testing, development and maintenance. For this project, we are following the steps of the Waterfall Model only up to the testing phase.

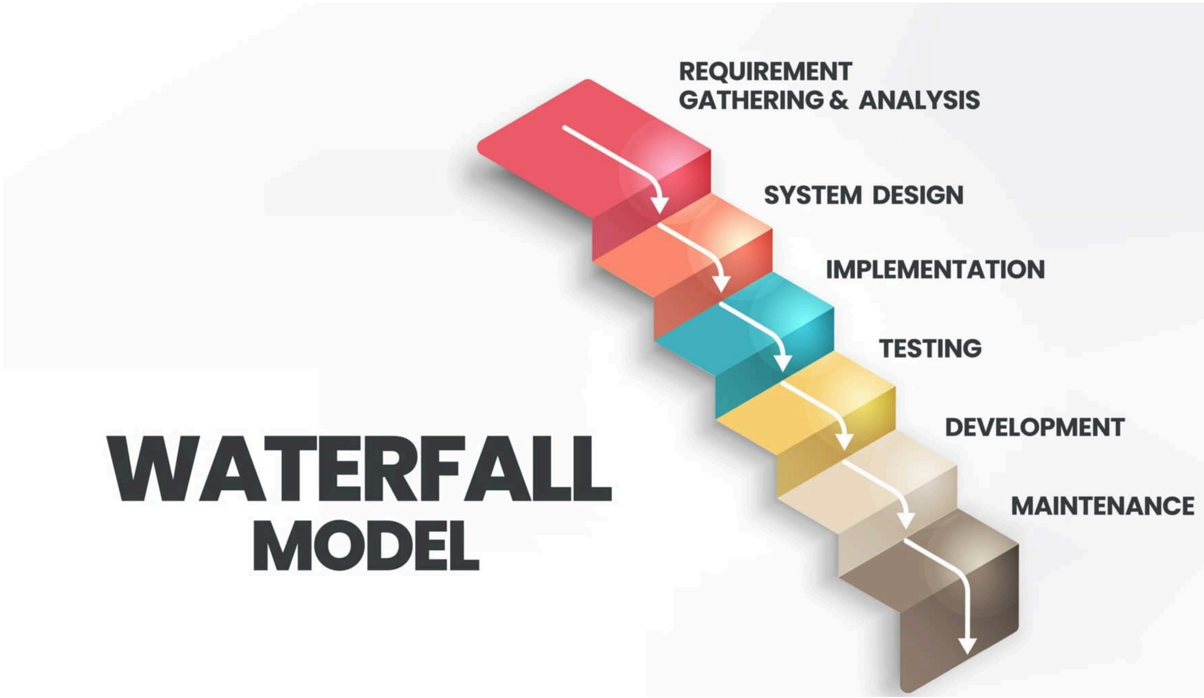

Figure 2.1 : Waterfall Model

Details Activities

Phases	Detail Activities	Description
Planning	● Group meeting ● Task distribution ● Project proposal writing	Sets up the project to achieve team goals, assigning tasks and creating a project proposal according to the objectives, scope and ensuring to fulfill the KADA requirements.
Analysis and Design	● Requirement Gathering ● SRS writing ● Use Case processes ● Database Design ● SDD writing	Gather and document KADA requirements in SRS, define user interactions with use cases and design the database using ERD. SDD is then created to provide technical architecture and clear specifications for building the system.
Implementation	● Development Tools Setup ● Integrating code ● Review code implementation	Preparing the suitable software and tools for coding, then combining the code to ensure they can work together and finally reviewing the code to check there's no error and can function properly.
Testing	● Test Planning ● User Acceptance Testing (UAT) ● Integration Testing ● STD writing	Develop a testing plan and involve real users to evaluate the software, ensuring it meets KADA requirements. Integration testing will verify the various part of system function and writing STD help to maintain the organization in testing approach.

Project Deliverables

Phases	Deliverables	Duration (week)
Planning	Project Proposal	2
Analysis	Software Requirements Specification (SRS)	2
Design	Software Design Document (SDD), Entity Relationship Diagram (ERD)	2
Implementation	PHP, MySQL	3
Testing	Software Test Description (STD)	5

2.3 Project Duration

The project aims to include all the information and activities that are required to develop the KADA Cooperative system. Figure 2.2 shows four major activities: Planning, Analysis and Design, Implementation and Testing. The assumption of the project to be completed in about 14 weeks, including 5 days a week.

The project will begin with a planning phase during the first week, which will involve a project meeting and task distribution among project members. The project proposal is the deliverable of this phase. The project then moves on to the analysis and design phase, where the system's requirements are gathered in a week and the software and database design of the system are designed and documented. The Software Requirements Specification (SRS) for the requirement and the Software Design Document (SDD), which will be revised at each design stage, are therefore the deliverables in this phase.

Following that, the system's implementation phase will be carried out. In this phase, the development tools for the system's development will be determined, and each person in charge will write a process. All written code will be combined to create a complete KADA Cooperative system once each step has been implemented. Lastly, at the end of the project, the testing phase will be carried out step-by-step when the system has been fully implemented. Following that, an iteration testing phase for the entire system will be conducted to continue the testing. The System Testing Documentation (STD) is the final outcome of the testing.

KADA Cooperative System

KodeLab

Project start: Sun, 10/13/2024

Display week: 1

SIMPLE GANTT CHART by Vertex42.com
https://www.vertex42.com/ExcelTemplates/simple-gantt-chart.html

TASK	ASSIGNED TO	START	END
Planning			
Group Meeting	Gokce Aslan	10/13/24	10/15/24
Task Distribution	Hayden Cook	10/15/24	10/17/24
Project Proposal Writing	Jens Martensson	10/18/24	10/25/24
Analysis and Design			
Requirement Gathering	All Members	10/26/24	10/31/24
SRS Writing	All Members	10/31/24	11/5/24
Use Case Process 1: Login	Aflah	11/4/24	11/16/24
Use Case Process 2: Member Registration	Adriana	11/4/24	11/16/24
Use Case Process 3: Loan Application	Fizana	11/4/24	11/16/24
Use Case Process 4: Member Statement	Pravin	11/4/24	11/16/24
Use Case Process 5: Board of Director Report	Adriana	11/4/24	11/16/24
Use Case Process 6: Admin Management	Fizana	11/4/24	11/16/24
Use Case Process 7: Review Application	Haani	11/4/24	11/16/24
Database Design	Haani, Aflah, Pravin	11/9/24	11/17/24
SDD Writing	All Members	11/16/24	11/23/24
Implementation			
Development Tools Setup	All Members	11/23/24	11/28/24
Process 1: Login	Aflah	11/26/24	12/8/24
Process 2: Member Registration	Adriana	11/27/24	12/9/24
Process 3: Loan Application	Fizana	11/27/24	12/9/24
Process 4: Member Statement	Pravin	11/27/24	12/9/24
Process 5: Board of Director Report	Adriana	11/27/24	12/9/24
Process 6: Admin Management	Fizana	11/27/24	12/9/24
Process 7: Review Application	Haani	11/27/24	12/9/24
Integrating Code	All Members	12/4/24	12/11/24
Review Code Implementation	All Members	12/11/24	12/18/24
Testing			
Test Planning	Gokce Aslan	12/14/24	12/18/24
Unit Testing : Login	Hayden Cook	12/18/24	12/25/24
Unit Testing : Member Registration	Jens Martensson	12/20/24	12/28/24
Unit Testing : Loan Application	Nuria Acevedo	12/21/24	12/28/24
Unit Testing : Member Statement	Olivia Wilson	12/24/24	12/31/24
Unit Testing : Board of Director Report	Olivia Wilson	12/25/24	1/1/25
Unit Testing : Admin Management	Olivia Wilson	12/29/24	1/5/25
Unit Testing : Review Application	Olivia Wilson	12/28/24	1/5/25
Iteration Testing	Olivia Wilson	12/31/24	1/15/25
STD Writing	Olivia Wilson	1/1/25	1/16/25

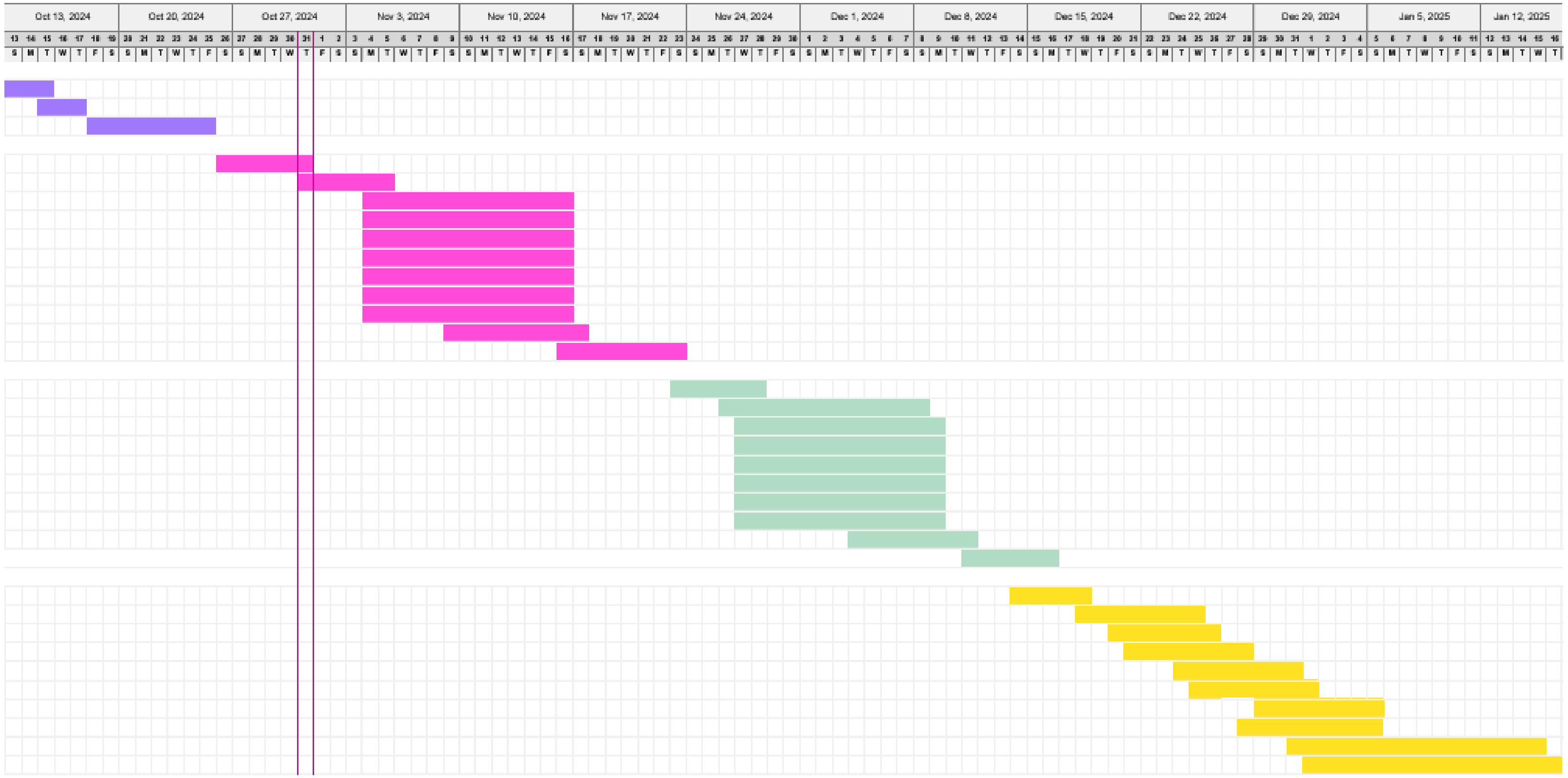

Figure 2.2 : Gantt Chart